

Diabetes Risk Factors

in the Pima Indian Population

A statistical analysis using descriptive statistics, inferential testing, and regression modelling



Unit 12 · MSc Artificial Intelligence · University of Essex

Dataset: Pima Indian Diabetes Database (Kaggle)

Smith et al. (1988); Khare, A.D. (2022)

Presentation Structure

1

Introduction & Dataset

768 Pima Indian women · 9 variables · diabetes outcomes

2

Data Exploration

Distributions, descriptive statistics, outliers

3

Diabetes Rates by Subgroup

BMI, age, and pregnancy categories

4

Inferential Statistics

Hypothesis tests, correlation, and regression models

5

Model Evaluation

Logistic regression performance & interaction effects

6

Conclusions & Recommendations

Findings, limitations, and six recommendations



Section 1

Introduction & Dataset



Dataset Overview: Pima Indian Diabetes Database

Study Population

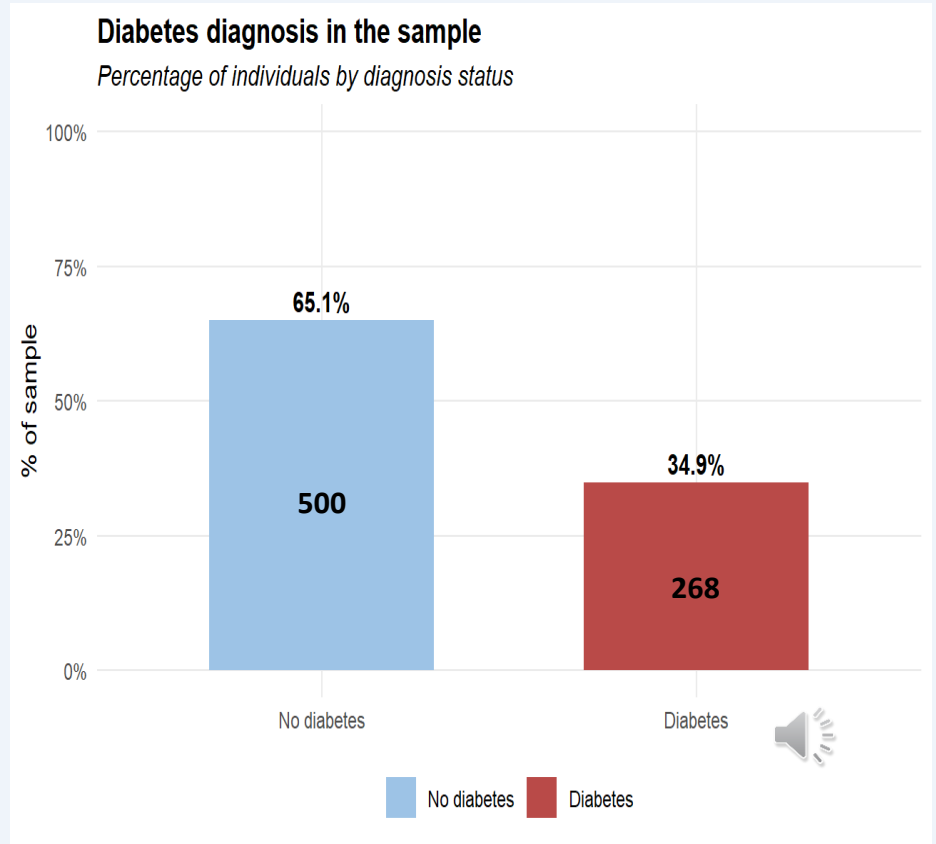
768 Pima Indian women aged 21–81
from the Phoenix, Arizona area

34.9% diabetes prevalence: 3–4× the global average (~10%)

*Source: National Institute of Diabetes and Digestive and Kidney Diseases
(Smith et al., 1988; Khare, 2022; International Diabetes Federation, 2021)*

Variables

- Pregnancies · Glucose · Blood Pressure
- Skin Thickness · Insulin · BMI
- Diabetes Pedigree Function · Age
- Outcome (diabetes diagnosis: 0/1)



Section 2

Data Exploration & Descriptive Statistics



Age Distribution

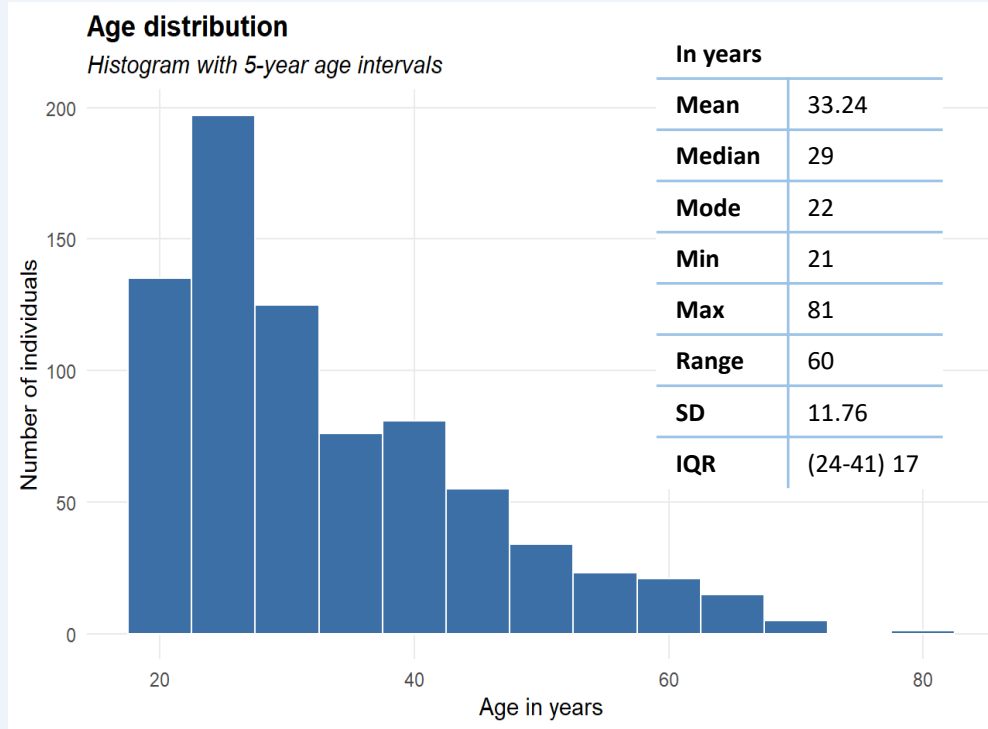


Fig 1. Age histogram (5-year intervals)

Median age: 29 years · IQR: 24–41 years

Sample is young-skewed, right-tailed distribution

Upper outliers: individuals above the 66.5 fence retained (biologically plausible)

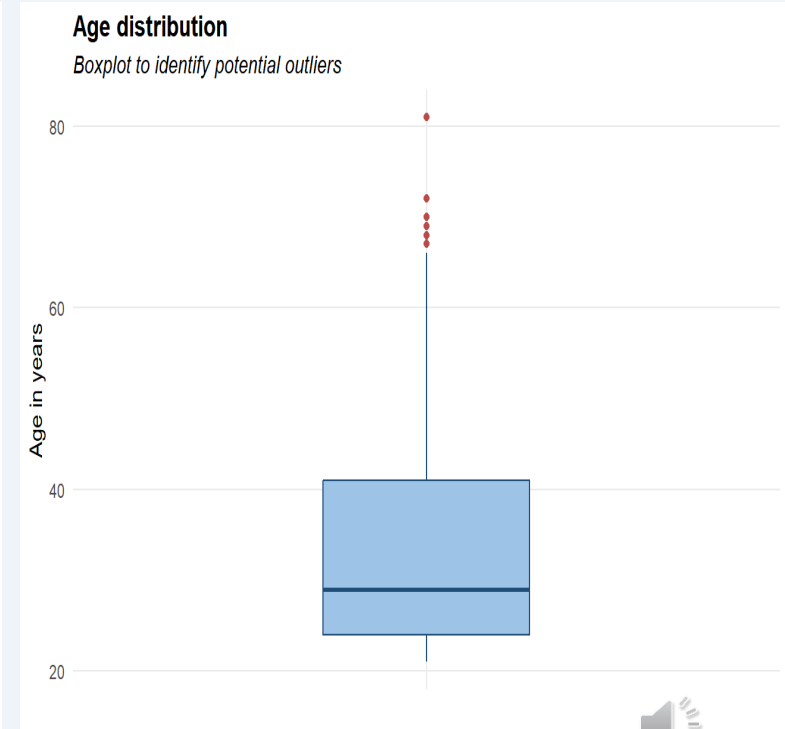


Fig 2. Age boxplot (outliers flagged)



Glucose & Blood Pressure Distributions

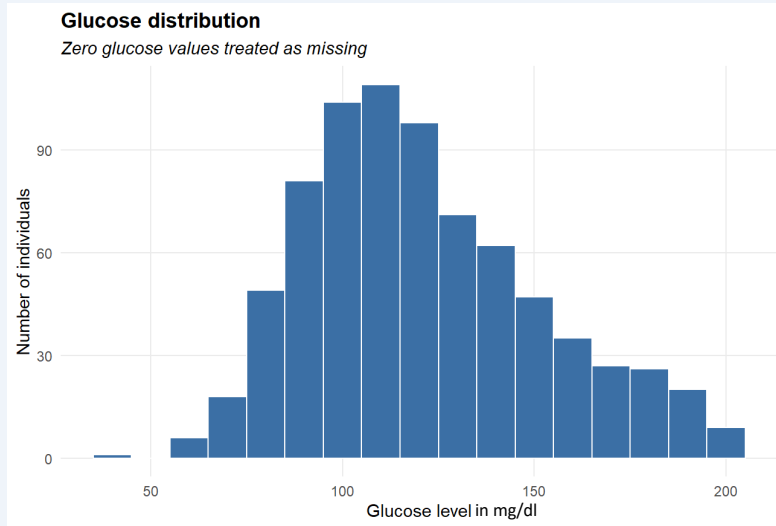


Fig 3. Glucose distribution (zero values = missing)

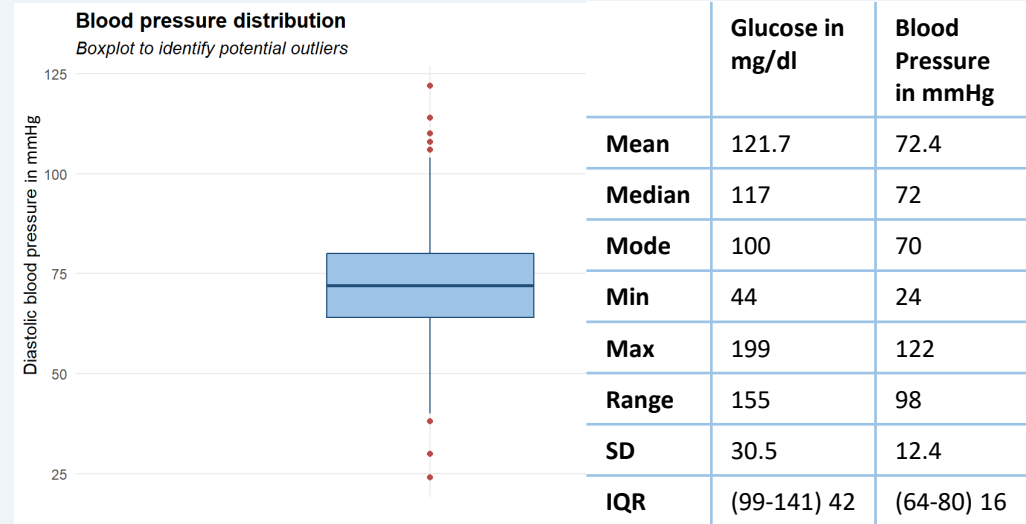


Fig 4. Blood pressure distribution (zero = missing)

Zero values in glucose, blood pressure, BMI, skin thickness, and insulin
treated as physiologically implausible missing data, excluded from regression sample
Regression n reduced from 768 → 392 (complete-case analysis)



BMI Distribution

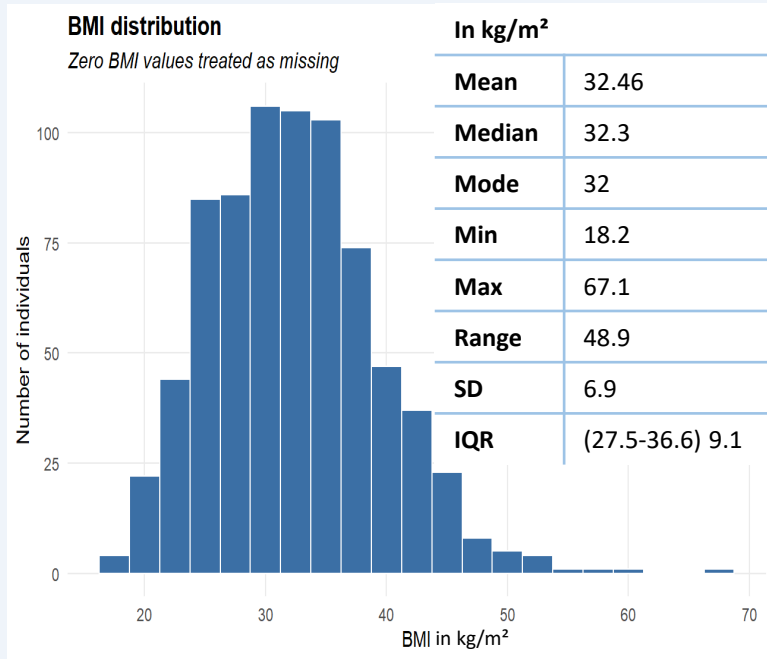


Fig 5. BMI histogram (zero values removed)

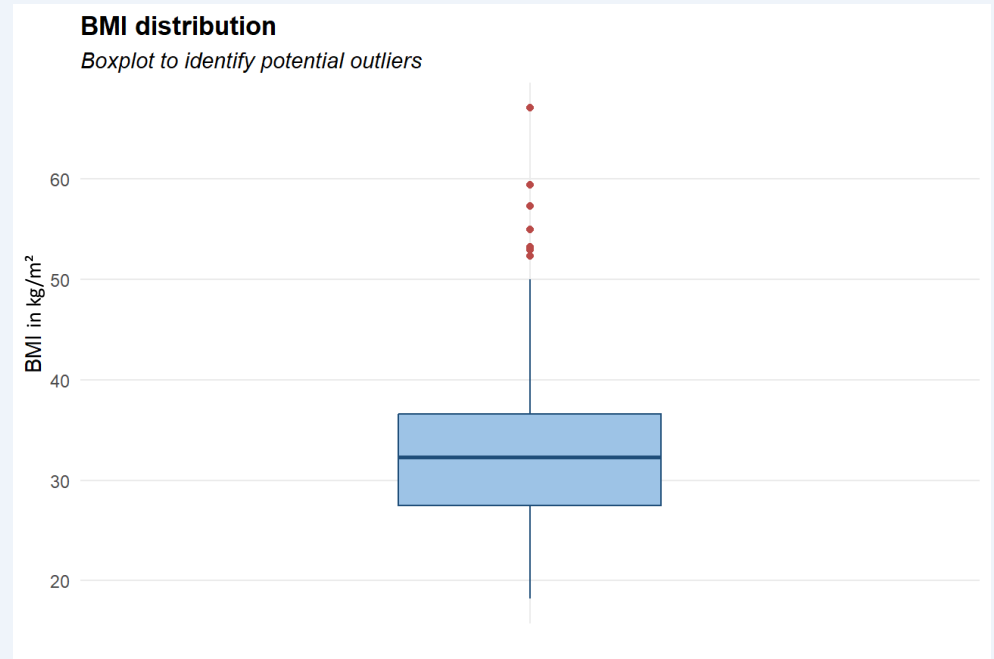


Fig 6. BMI boxplot, outliers up to 67.1



Median BMI: 32.3 · IQR: 27.5–36.6 · Upper outliers retained (up to 67.1)

Sample skewed toward overweight/obese range, relevant to classification model performance

Pregnancy Distribution

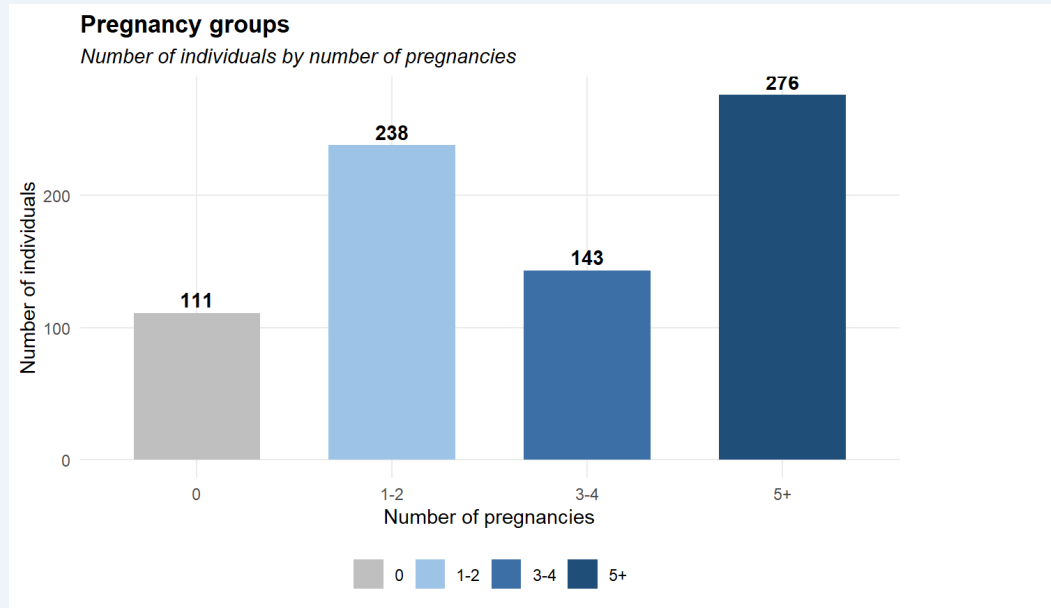


Fig 7. Pregnancy groups distribution

Nbr of pregnancies	Full sample (n=768)	Ever-pregnant only (n=657)
Mean	3.8	4.49
Median	3	4
Mode	1	1
Min	0	1
Max	17	17
Range	17	16
SD	3.4	3.22
IQR	(1-6) 5	-



Pregnancies, diabetic vs non-diabetic: 4.87 vs 3.30 · Wilcoxon W = 50985, $p = 3.7 \times 10^{-8}$ -> significant difference

Correlation Matrix: Continuous Variables



Correlation matrix

Pairwise Pearson correlations between continuous variables

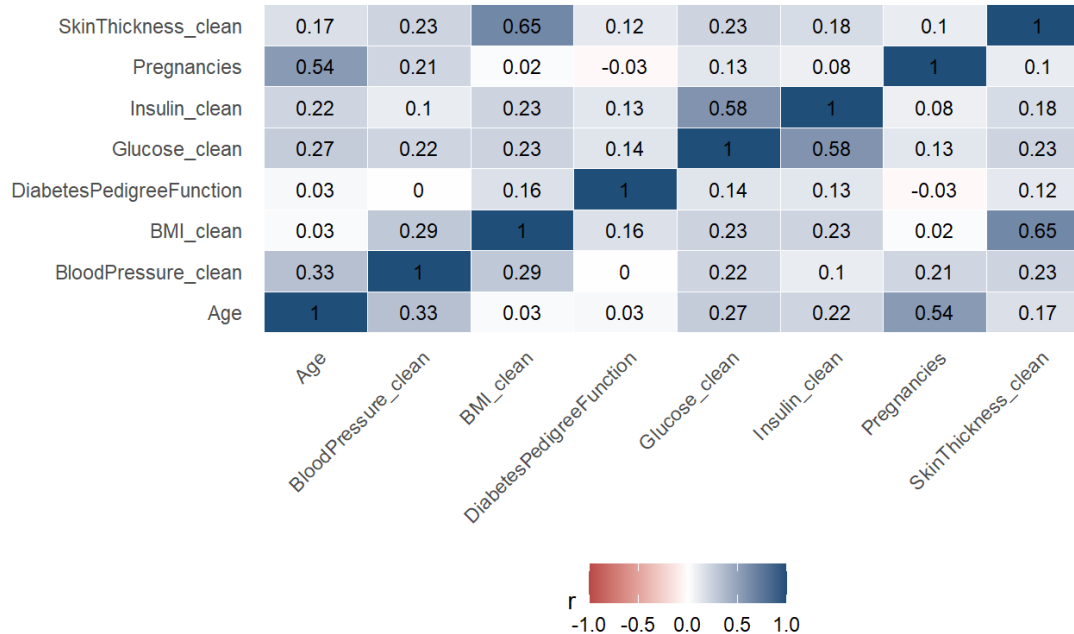


Fig 8. Pairwise Pearson correlations

Key Correlations

BMI ↔ Skin Thickness **r = 0.65**

Insulin ↔ Glucose **r = 0.58**

Age ↔ Pregnancies **r = 0.54**

Age ↔ Blood Pressure **r = 0.33**

BMI ↔ Blood Pressure **r = 0.29**

Note: No pair severe enough to destabilise regression, but BMI/SkinThickness ($r=0.65$) adds coefficient uncertainty

Section 3

Diabetes Rates by Subgroup

Diabetes Rate by BMI Category

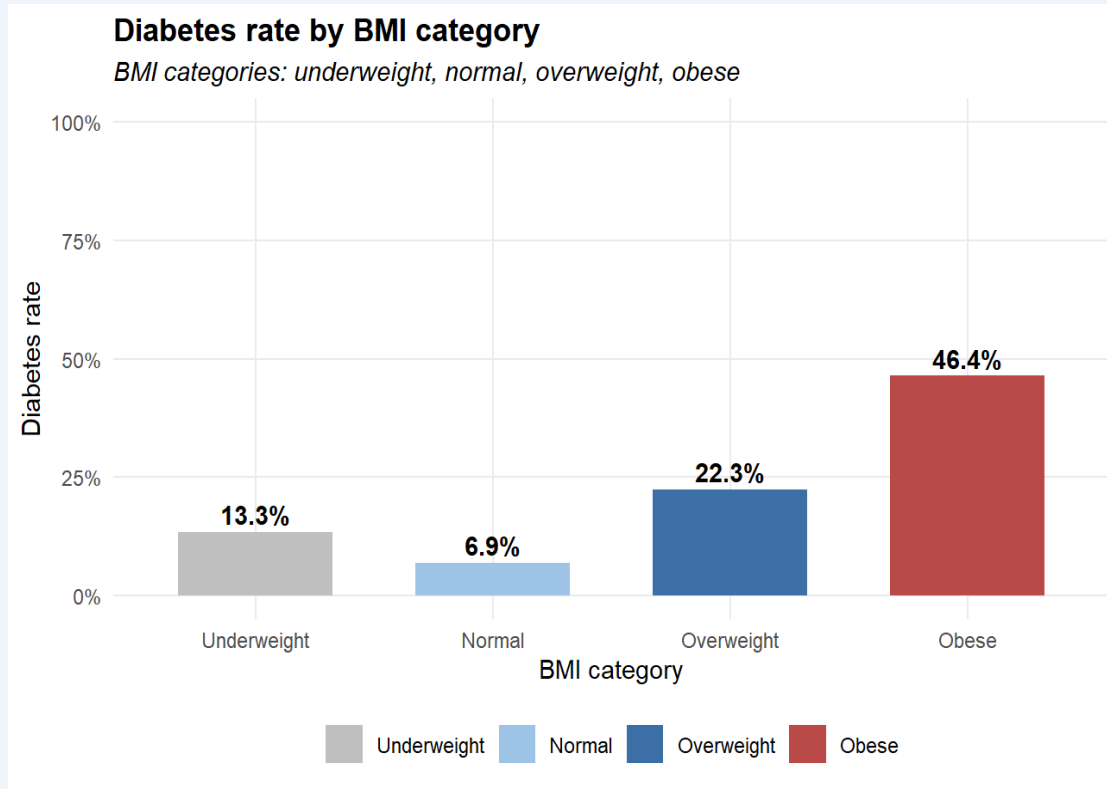


Fig 9. $\chi^2 = 78.25$, $p < 2.2 \times 10^{-16}$ (Pearson chi-square)

Dose-Response Gradient

6.9%

Normal BMI

13.3%

Underweight

22.3%

Overweight

46.4%

Obese



Strong dose-response association supports BMI as an independent predictor of diabetes risk (OR = 1.09, $p < 4.3 \times 10^{-10}$)

Diabetes Rate by Age Group

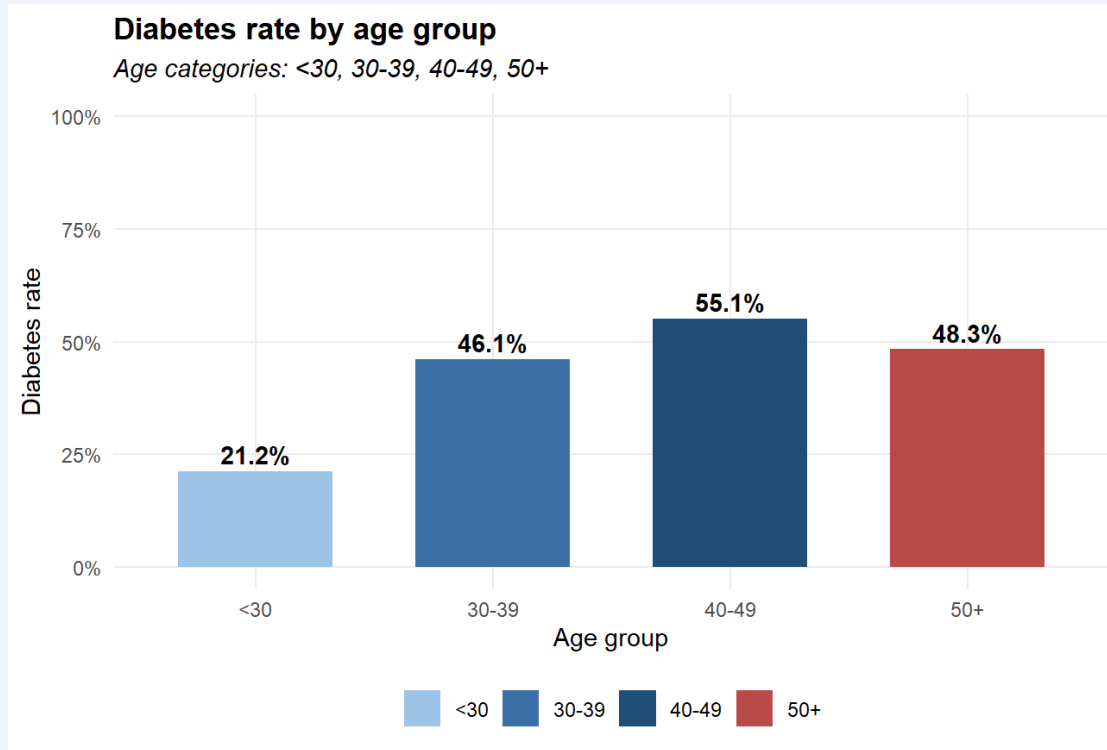


Fig 10. $\chi^2 = 69.92$, $p = 4.5 \times 10^{-15}$ (Pearson chi-square)

Age as a Risk Escalator

21.2%

Under 30

46.1%

30-39

55.1%

40-49

48.3%

50+



⚠ Plateau between 40-49 and 50+:
relationship is not strictly linear across full range

Glucose Levels by Diabetes Status

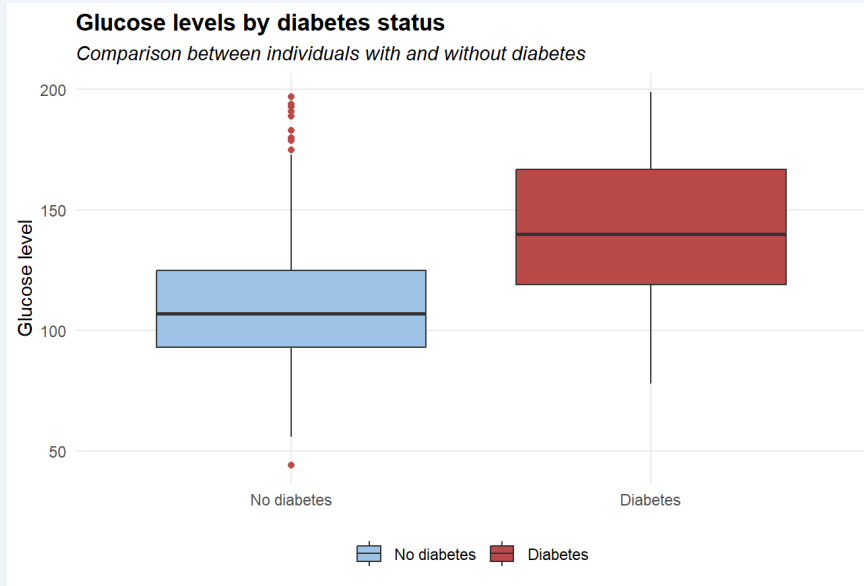


Fig 11. Glucose boxplot by diabetes status

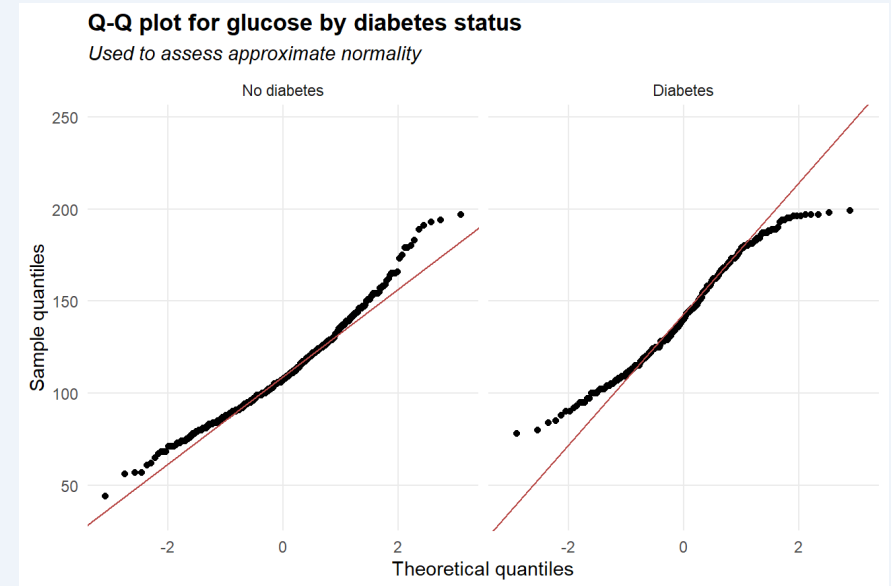


Fig 12. Q-Q plots assessing normality assumption

Diabetic mean: 142.3 mg/dL vs Non-diabetic mean: 110.6 mg/dL

Welch t-test: $t = -14.884$, $p < 2.2 \times 10^{-16}$ · Wilcoxon rank-sum: $p < 2.2 \times 10^{-16}$



Diabetes Rate by Number of Pregnancies

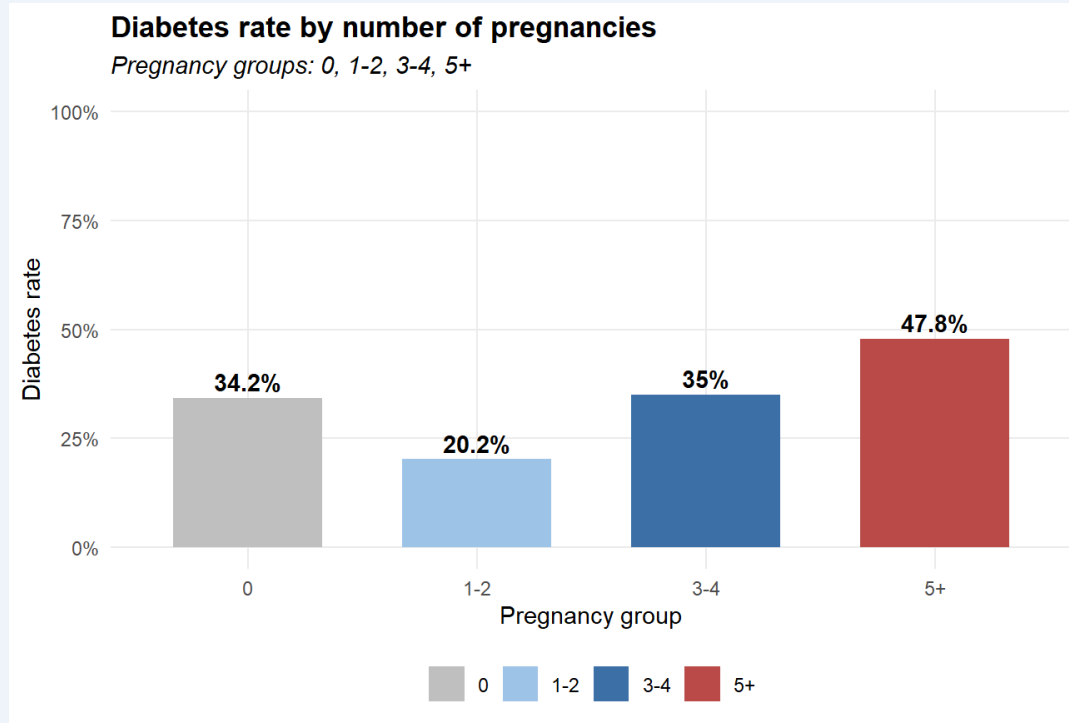


Fig 13. Diabetes rate by pregnancy group

Diabetes Rate by Pregnancy Group

0 → 34.2%
1-2 → 20.2%
3-4 → 35.0%
5+ → 47.8%

Steady rise from 1-2 onward, as expected.
Zero-pregnancy group breaks the pattern
Likely reverse causation: undiagnosed insulin resistance may suppress fertility

*Cross-sectional design precludes causal confirmation
(Dabelea et al., 2000)*



Section 4

Inferential Statistics & Regression Models

Glucose Levels Across Age Groups

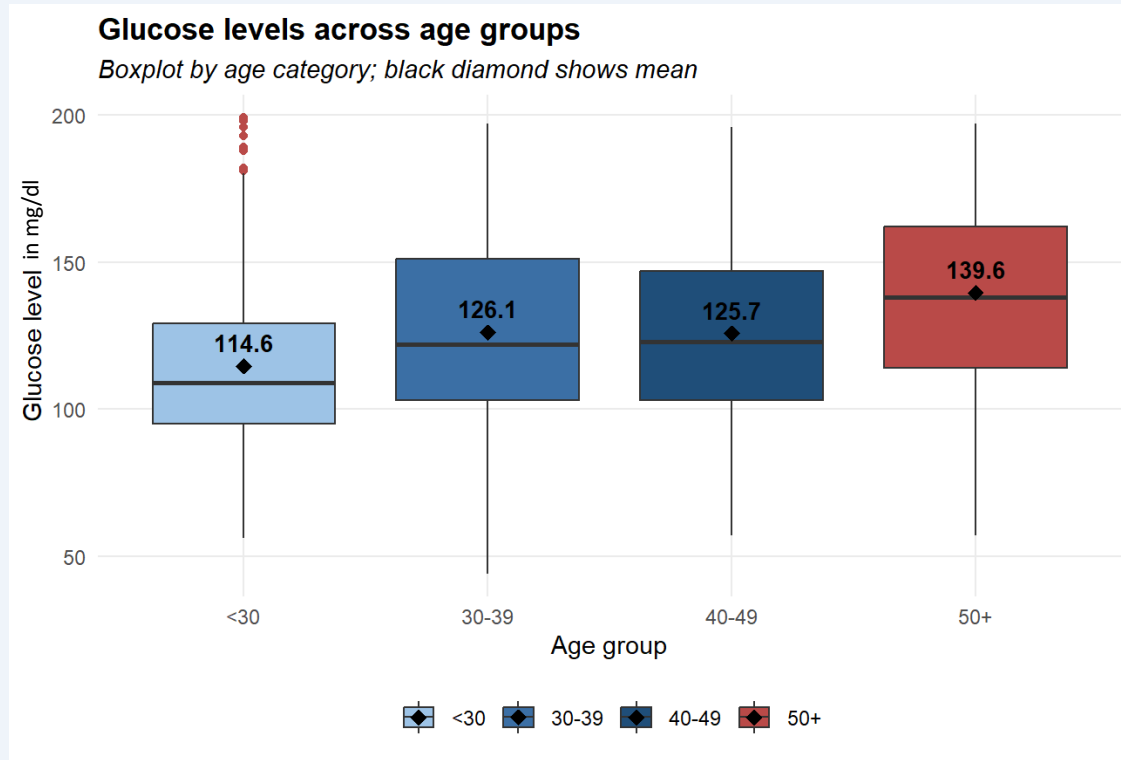


Fig 14. Glucose boxplot by age group

Findings

Mean Glucose by Age Group (mg/dL):

114.6

126.1

125.7

139.6

ANOVA $F(3, 759) = 20.44, p < 0.001$

Kruskal-Wallis: $H(3) = 59.04, p = 9.4 \times 10^{-13}$

(non-parametric robustness check: upper-tail deviation in <30 group)

Tukey HSD: 30–39 and 40–49 not significantly different from each other

50+ significantly higher than all other groups

<30 group: several high outliers (insulin resistance?)

Multiple Linear Regression: Predicting Glucose

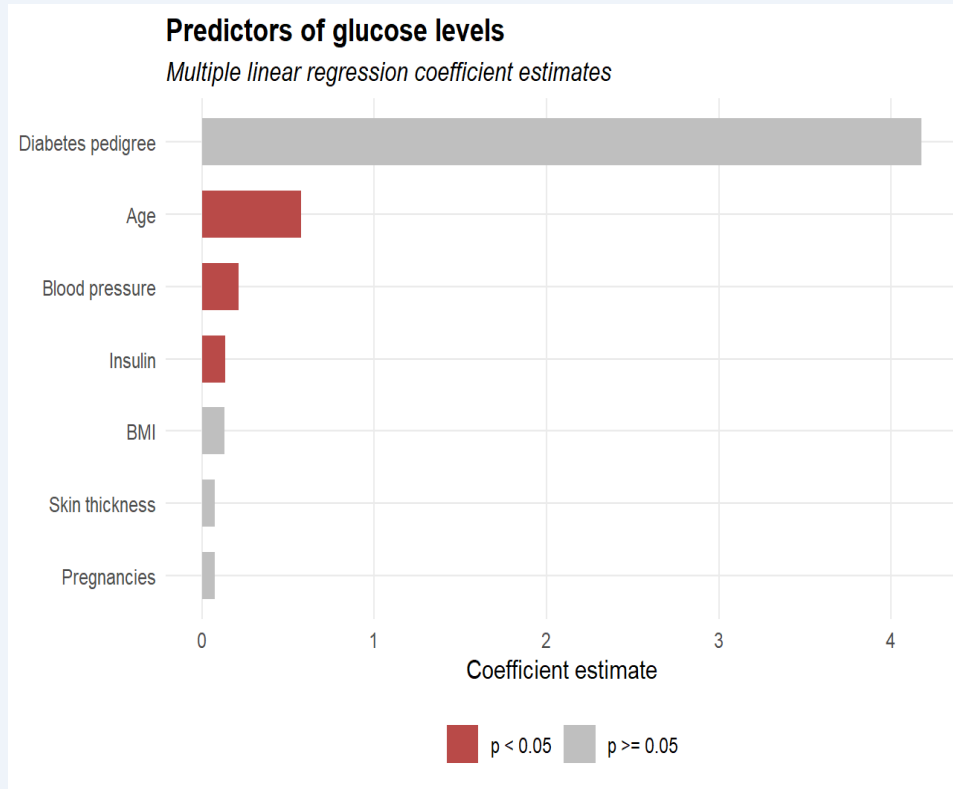


Fig 15. Coefficient estimates (red = $p < 0.05$)

Model Performance

Adjusted $R^2 = 0.39$

39% of glucose variance explained

$F(7,384) = 36.76$

$p < 2.2 \times 10^{-16}$ overall model significant

Significant Predictors ($p < 0.05$)

Insulin: strongest physiological predictor

Age: independent life-course effect

Blood pressure - also significant ($p < 0.05$)

⚠ **Diabetes pedigree:** large raw coefficient (~4.2) but $p \geq 0.05$
Scaling artefact, unitless score not comparable to physiological measurements without standardisation

Linear Regression: Diagnostic Plots

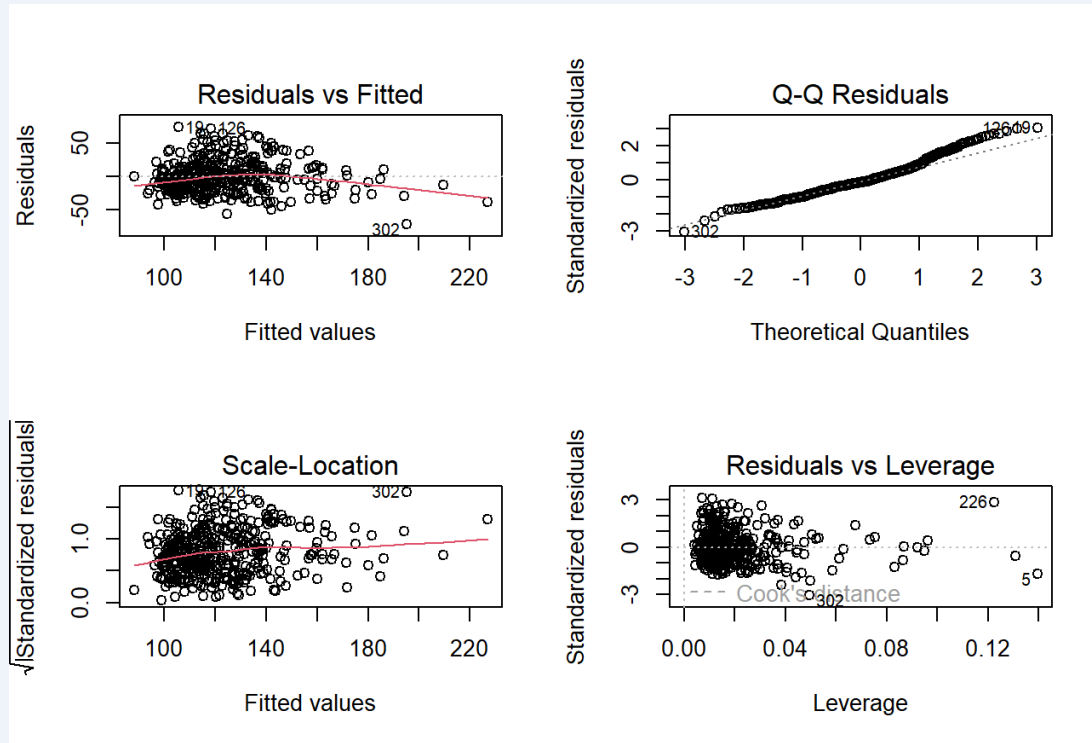


Fig 16. Standard diagnostic plot panel (R base graphics)

Diagnostic Findings



Residuals vs Fitted

Mild systematic pattern, non-linearity possible



Q-Q Residuals

Acceptable normality with upper-tail deviation



Scale-Location

Mild upward trend → non-constant error variance at high fitted values



Leverage

Obs. 226 & 5 have elevated leverage, no Cook's Distance breach



Linear Regression: Actual vs Predicted Glucose

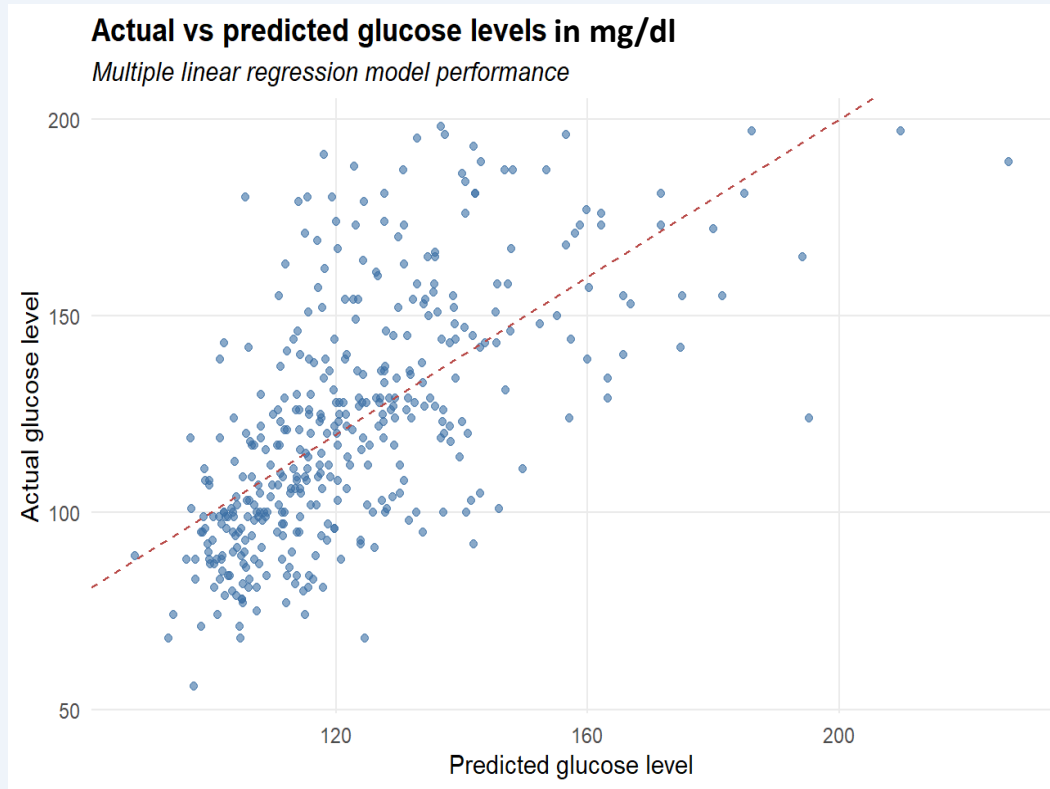


Fig 17. Predicted vs actual glucose (dashed = perfect prediction)

Model Interpretation

Purpose:

Explain glucose mechanism, not clinical prediction

Wide scatter

60% of variance unaccounted for

Missing predictors likely include:

- Diet & physical activity
- HbA1c & fasting status
- Diabetes duration



Section 5

Logistic Regression & Model Evaluation

Logistic Regression: Three-Predictor Model (BMI, Age, Glucose)

Odds Ratio Interpretation

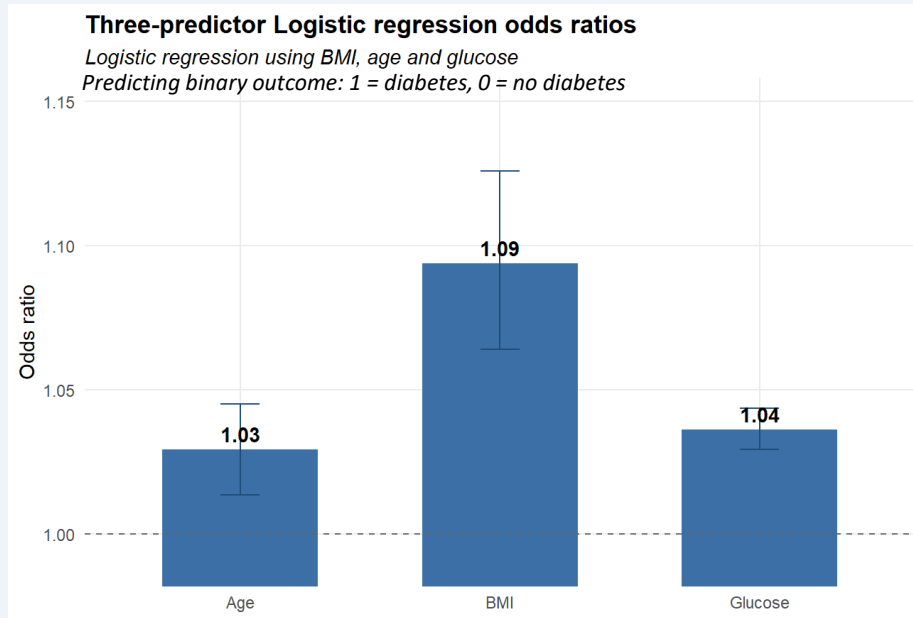


Fig 18. Three-predictor odds ratios



BMI

OR = 1.09

95% CI: 1.06–1.13

Each unit $\uparrow$ = 9% higher odds of diabetes

$p < 4.3 \times 10^{-10}$



Glucose

OR = 1.04

95% CI: 1.03–1.04

Each mg/dL $\uparrow$ = 4% higher odds

$p < 2 \times 10^{-16}$



Age

OR = 1.03

95% CI: 1.01–1.05

Each year $\uparrow$ = 3% higher odds, significant in this model

$p = 0.000238$



Model 1: Logistic Regression: Four-Predictor Model (BMI, Age, Glucose, Pregnancies)

Odds Ratio Interpretation

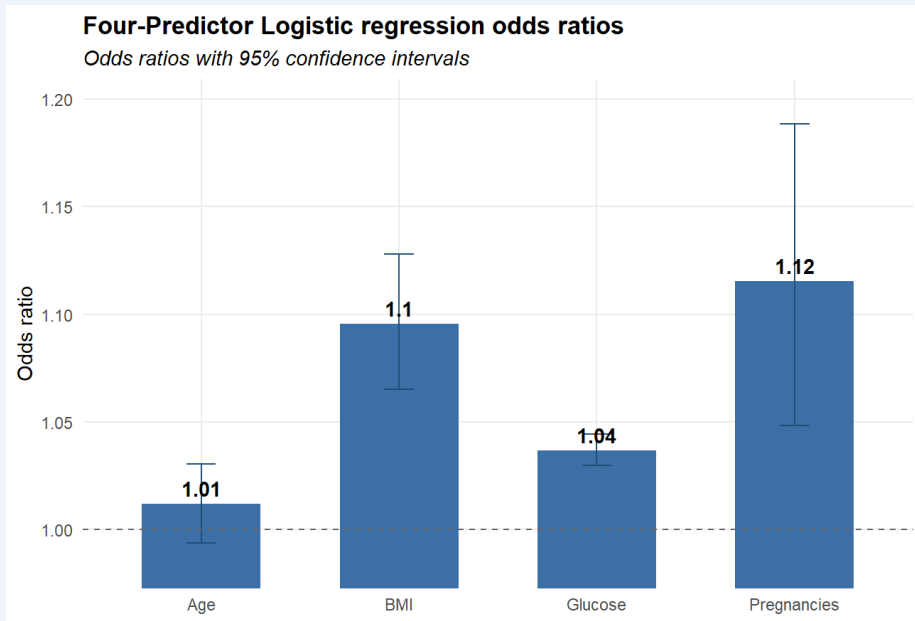


Fig 19. Odds ratios with 95% confidence intervals



BMI

OR = 1.10

95% CI: 1.07–1.13

Each unit $\uparrow$ = 10% higher odds of diabetes

$p < 4.6 \times 10^{-10}$



Glucose

OR = 1.04

95% CI: 1.03–1.04

Each mg/dL $\uparrow$ = 4% higher odds

$p < 2 \times 10^{-16}$



Age

OR = 1.01

95% CI: 0.99–1.03

Each year $\uparrow$ = 1% higher odds, not statistically significant

$p = 0.194$



Pregnancies

OR = 1.12

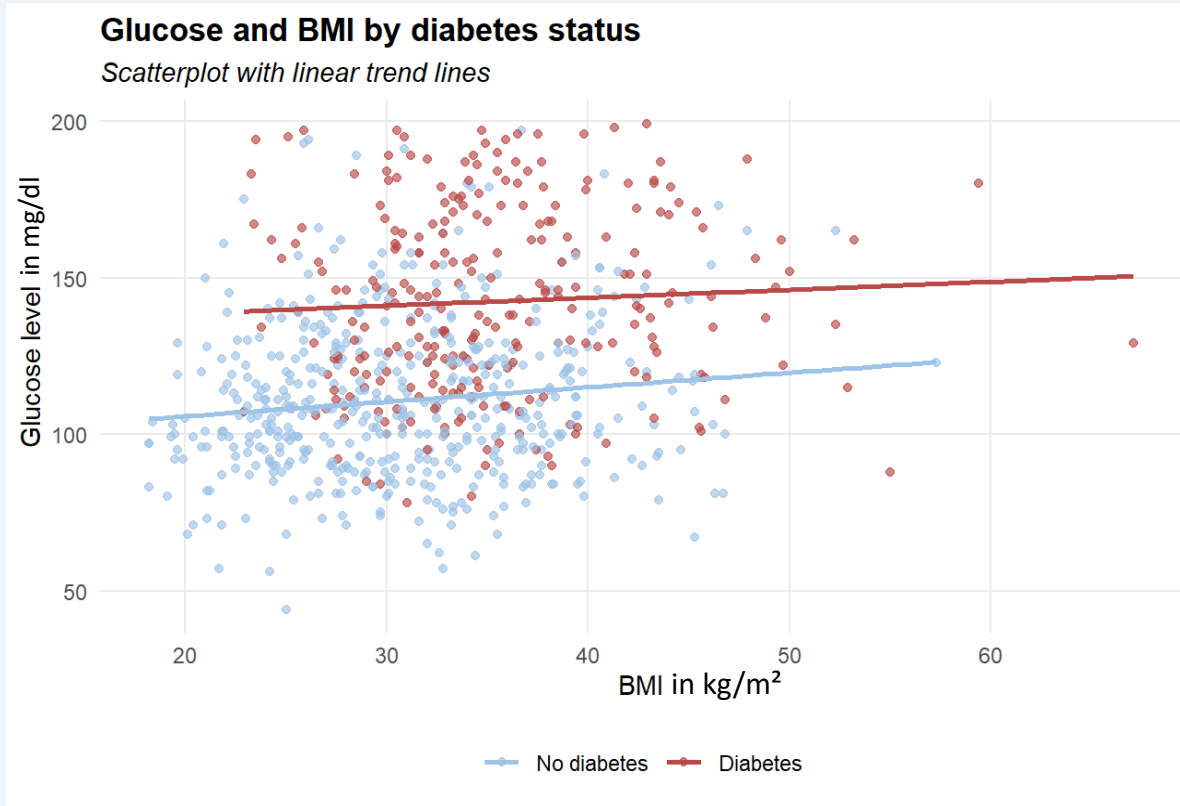
95% CI: 1.05–1.19

Each additional pregnancy = 12% higher odds

$p < 0.001$



Glucose and BMI: Joint Relationship by Diabetes Status



Key Observation

Same BMI → diabetics sit higher on glucose

Glucose & BMI carry **non-redundant information**

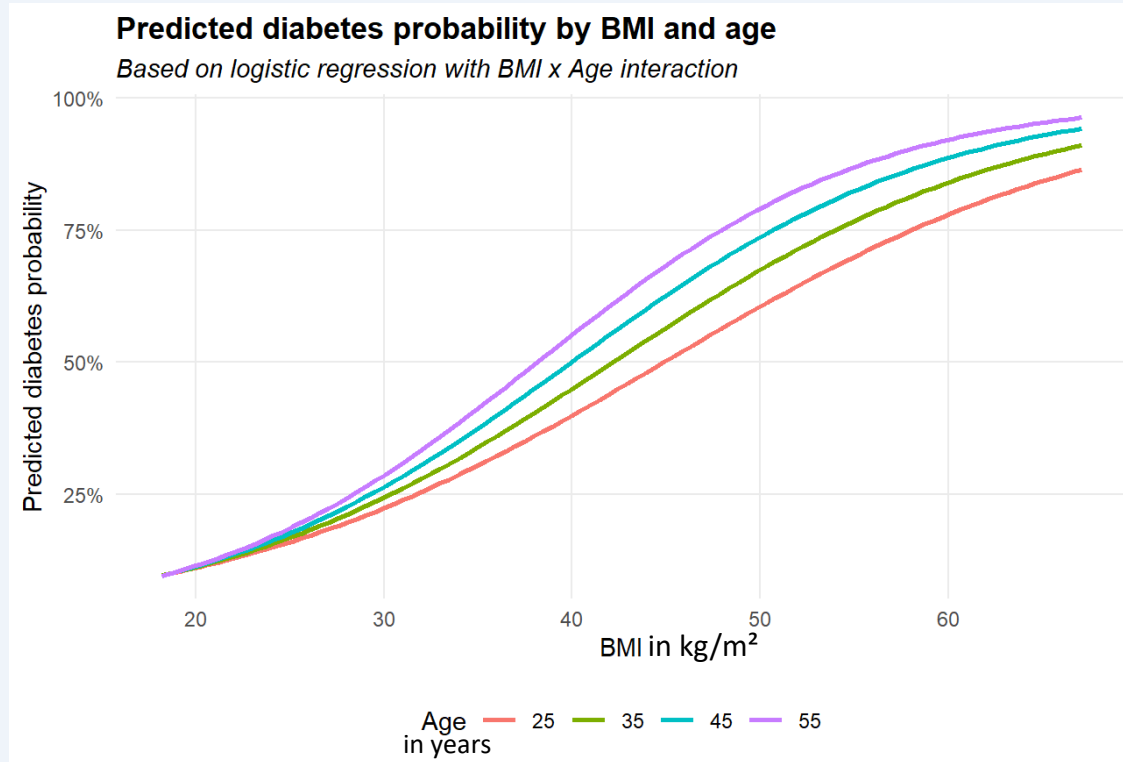
Both should be considered in diabetes risk screening models.

(Supports Recommendation 1)



Fig 20. Scatterplot with linear trend lines by diagnosis status

Model 2: Logistic Regression: BMI × Age Interaction



Interaction Analysis

LRT Result:

Δ deviance = 0.564, $p = 0.453 \rightarrow$ NOT significant

Interaction OR = 1.001

95% CI spans 1

Without interaction (Model 1)

AIC = 722.9 ✓ preferred

With interaction (Model 2)

AIC = 724.3

Curves widen visibly at BMI > 35

$\rightarrow$ power limitation, not a true null



Fig 21. Logistic regression with BMI × Age interaction (van de Schoot et al., 2021)

Logistic Regression: Classification Performance

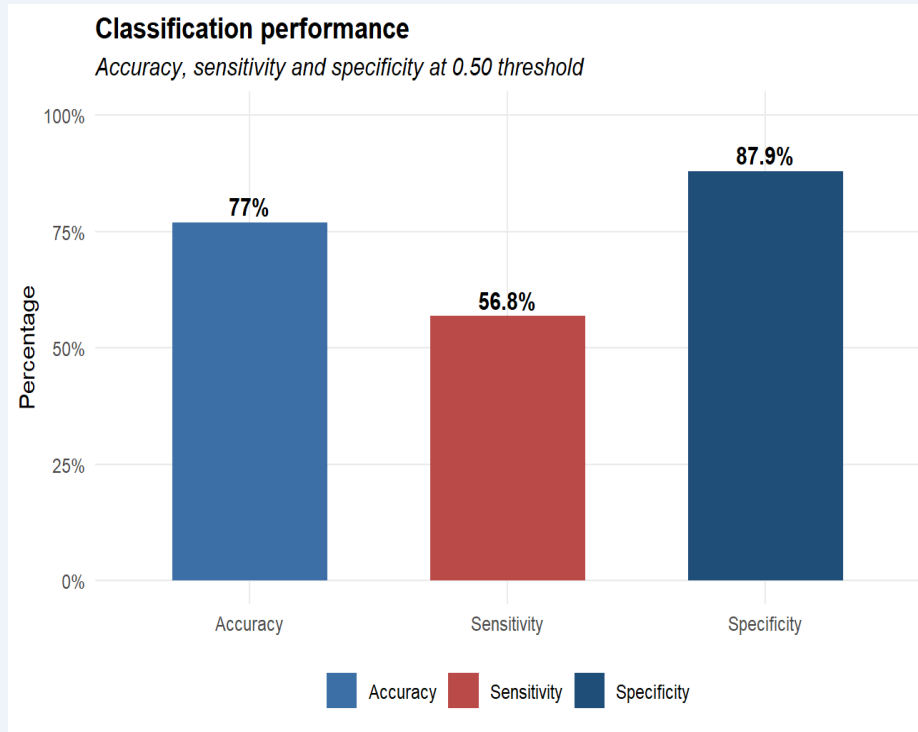


Fig 22. Performance at 0.50 threshold

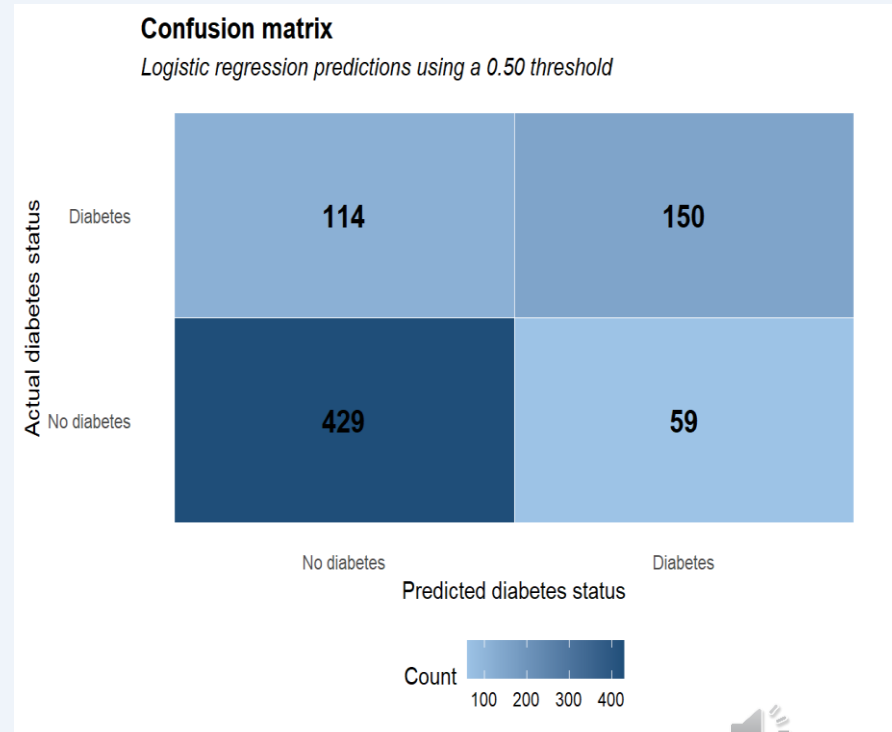


Fig 23. Confusion matrix (logistic regression)

77.0% accuracy · 87.9% specificity · **56.8% sensitivity** ⚠ · Hosmer-Lemeshow: $p = 0.197$ (adequate fit) (Hosmer et al., 2013)
114 true diabetics misclassified as negative vs 59 false positives → 2:1 false negative ratio

Section 6

Conclusions & Recommendations

Summary of Key Findings

Glucose



Mean 142.3 vs 110.6 mg/dL · OR = 1.04 · $p < 2 \times 10^{-16}$

Insulin



Strongest in linear regression · mechanistic role, not classificatory

BMI



6.9% → 46.4% dose-response · OR = 1.09 · independent of glucose

Pregnancies



34.2% (0 preg.) vs 20.2% (1–2) · likely reverse causation

Age



21.2% (<30 years) → 55.1% (40–49 years, peak) · significance not stable across model

MLR model



Adjusted $R^2 = 0.39$ · 60% variance unexplained



Critical Appraisal: Limitations

Missing Data



768 → 392 · exclusion not confirmed random · selection bias possible

External Validity



No train-test split · in-sample figures likely optimistic

Non-constant error variance



Scale-Location: mild upward trend · SEs underestimated at high glucose

Population Specificity



Pima prevalence ~35% vs ~10% global · recalibration required

Collinearity



BMI–Skin $r=0.65$ · Insulin–Glucose $r=0.58$ · coefficient uncertainty understated

Statistical vs Clinical



$p < 0.05 \neq$ clinical fit · 56.8% sensitivity · 0.50 threshold inappropriate



Six Recommendations

1

Prioritise Glucose & BMI

Consistent, independent, non-redundant · obligatory screening inputs

2

Reconsider 0.50 Threshold

Try 0.30–0.35 · reflect clinical cost of missed diagnosis vs false referral

3

Age & Pregnancy as Context

Significant but not modifiable · contextual markers, not causal claims

4

Multiple Imputation

Replace complete-case analysis · sensitivity analysis to bound residual bias

5

External Validation

No train-test split applied · cross-validation is minimum requirement

6

Extend to Bayesian Methods

Informative priors justified · posterior handles class imbalance explicitly



Core Takeaway

Glucose and BMI are the most consistent predictors of diabetes risk. Age is significant in the simpler model, but loses significance once pregnancy history is included

The logistic model is adequate for population-level risk stratification.

Sensitivity limitations, missing data, absence of external validation, and non-constant error variance are conditions, not footnotes.

References

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